THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT



TANGA REGION

FORM FOUR MOCK EXAMINATION

CHEMISTRY 1 MARKING SCHEME

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
C	C	C	A	A	B	C	D	C	B

1Marks @

2.

LIST A	i	ii	iii	iv	v	vi
LIST B	E	D	C	F	B	A

1Marks @

3. (a). $BaCl_2(aq) + Na_2SO_4(aq) \longrightarrow BaSO_4(s) + 2NaCl(aq)$

$Ba^{2+}(aq) + 2Cl^-(aq) + 2Na^+(aq) + SO_4^{2-}(aq) \quad BaSO_4(s) + 2Cl^-(aq) + 2Na^+(aq)$

$Ba^{2+}(aq) + SO_4^{2-}(aq) \longrightarrow BaSO_4(s)$

(b). There would be incomplete combustion due to shortage of oxygen resulting into production of poisonous chemicals such as carbon monoxide which is poisonous.

Equation $C(s) + O_2(g) \longrightarrow CO(g)$

(c). i. Fluorine

ii. Iodine

iii. Iodine

iv. Cl = 2:8:7

4. (a). Permanent hardness of water is that contain dissolved ions of calcium and magnesium in form of sulphate and chlorine while temporary hardness of water is that water which contain dissolve calcium and magnesium in form of bicarbonate.

(b). permanent hardness of water can be removed by chemical means. Forexample the addition of sodium carbonate as per equation below

$$CaSO_4(aq) + Na_2CO_3(aq) \longrightarrow CaCO_3(s) + Na_2SO_4(aq)$$

But temporally hardness of water can simply be removed by boiling. The equation below represent

$$CaHCO_3(aq) \xrightarrow{heat} CaCO_3(s) + H_2O(l) + CO_2(g)$$

(c). impact the hairs and skin
It waste more soap
Stain utensils and clothes
Lime scale formation in water boilers

5. (a). (i) alkane (ii) Alkene (iii) Alkanol or Alcohol
(b) Methane , Ethene , Methanol
(c) (i) $CH_2CH_2 \xrightarrow[Ni]{H_2} CH_3CH_3$

(ii) $CH_3CH_2OH \xrightarrow[180\ ^0C]{Conc.\ H_2SO_4} CH_2CH_2$

6 (a)

Symbols	Zn	S	O	H_2O
	22.2	11.6	22.3	43.9
	22.2/65	11.6/32	22.3/16	43.9/18
	0.34/0.34	0.36/0.34	1,4/0.34	2.44/0.34
	1	1	4	7

Empirical formula is $ZnSO_4.7H_2O$

M.F = $(ZnSO_4.7H_2O)n = 283$

$(ZnSO_4.7H_2O)n = 283$

$287n = 283$

n = 1

Therefore molecular formula is $ZnSO_4.7H_2O$

(b). tell us which compound react with another to form product

It tell us how many reactant and product formed

To perform calculations

(c). $2H_2 + O_2 \longrightarrow 2H_2O$

n= m/mr = 5/2 = 0.25

From the equation above 2mol of hydrogen = 2mol water

Then 0.25mol of H_2 = 0.25mol H_2O

It's mass will be n x Mr= 0.25m0l x 18g/mol

The mass of water will be 4.5g.

7. (a). $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

The role played by catalyst is to speed up the rate of formation of ammonia

(b). $2H_2S(g) + SO_2(g) \longrightarrow 2H_2O(l) + 3S(s)$

(c). i. Sodium/ Potassium.

ii. Calcium

iii. Lead.

8. (a). the number of proton is equal to the number of neutron which is given by 40-20 giving 20. The element x is a metal.

(b). (i). the temperature should be lowered while the pressure should be maximized

(ii). It will slow down the formation Sulphur trioxide as the main component of contact process.

(iii). The catalyst is vanadium pentaoxide (V_2O_5) or platinum Asbesto.

Section C

9. Parent materials

Climate

Organisms

Topography

Time

10. Cause soil erosion

    Loss of habitat to some organisms - acidic Rain, Reduction of area for settlement.

    Cause acid Mine drainage

    It results into lung diseases due prolonged exposure to coal dusts in mines

    It cause global warming

    Air Pollution,

11. (a). The first faraday's law states that the mass of element deposited at the electrode varies proportional to the quantity of electricity passed through the electrolytes while faraday's second law states that if the same amount of electricity is passed through different electrolytes, the mass of ions deposited at the electrode are direct proportional to the chemical equivalent. **(2marks)**

(b) Ionization of Silver at the Cathode

$Ag^{+}(aq) + e^{-} \longrightarrow Ag(s)$ **(0.5 marks)**

From 1F=96500C

X= 9650C (1marks)

By cross multiplication $\frac{= 1F \times 9650C}{96500c} = 0.1F$ **(1.5 marks)**

From the ionization equation 1F = 108g/mol **(1mark)**

0.1F = X

Mass of Ag = 0.1Fx 108g/mol/1F= 10.8g **(1.5marks)**

Volume of oxygen $4OH^{-}_{(aq)} \longrightarrow 2H_2O_{(l)} + O^{2}_{(g)} + 4^{e-}$ **(1 mark)**

4F= 22.4dm3 **(1mark)**

0.1F= X

By cross Multiplication

$\frac{0.1F \times 22.4dm^3}{4F} = 0.56dm^3$ or $560cm^3$ **(1.5 mark)**

Therefore the volume of oxygen gas is 0.56dm3 or 560cm^3.

(c). Electroplating with metals
Electrorefining of metals
Extraction of metals
Purification of metals
Production of chemicals (4marks)